

Research Question and Hypotheses: Physics-Grounded, Label-Efficient Measurement of Oral Soft Tissue in OCT

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Working title: *Physics-grounded, label-efficient learning for trustworthy measurement of oral mucosa in optical coherence tomography.*

In one sentence: we ask whether building the physics of OCT image formation into how a model learns lets it measure oral epithelium accurately from very few labelled scans, and attach statistically guaranteed uncertainty to every measurement so it can serve as the measurement layer of a clinical decision support system.

1. Problem statement and motivation

Clinical need. Many oral soft-tissue conditions announce themselves as changes in the layered structure of the mucosa: the epithelium thickens or thins, its boundary with the underlying connective tissue (lamina propria) blurs or breaks, and the scattering of the connective tissue changes. Today these changes are judged by eye and confirmed by biopsy. An objective, non-invasive measurement would help decide *where* to biopsy, whether a lesion is changing over time, and how inflamed or fibrotic tissue is.

Why OCT. OCT images the first 1–2 mm of oral mucosa at micrometre resolution without radiation. It clearly separates epithelium from lamina propria in vivo, and has been used to establish site-specific normal epithelial thickness — for example a mean of about 336 μm across sites, from about 100 μm on the floor of the mouth to about 660 μm on the cheek in 28 healthy volunteers ([Di Stasio et al., 2019](#)). OCT readings of oral dysplasia and malignancy have shown excellent agreement with histopathology in a 50-patient study ([Wilder-Smith et al., 2009](#)).

The obstacle. Interpreting OCT volumes manually is slow and requires expertise, so automated analysis is needed. But:

- there is **no public oral OCT dataset**, and each group's data is small (tens of patients);
- **pixel-level labels are expensive**, and abnormal cases with histology are rare;
- raw OCT brightness depends heavily on the **device**, focus, and acquisition, so models trained on one system may not transfer;

- a decision support system needs **measurements it can stand behind**, not just a best guess.

The opportunity. OCT image formation is physically well understood: light is attenuated with depth at a rate set by the tissue, and each layer has characteristic scattering. Existing oral OCT deep-learning pipelines learn purely from labelled examples. If that physics is built into *how the model learns*, the model may need far fewer labels, generalise better across devices, and produce physically meaningful outputs — and if those outputs carry guaranteed uncertainty, they become usable for clinical decisions.

2. Related work and the gap

Oral OCT analysis

- **Clinical feasibility.** In vivo OCT distinguishes epithelium, basement membrane, and lamina propria and supports diagnosis of dysplasia and malignancy ([Wilder-Smith et al., 2009](#)); normative site-specific epithelial thickness has been established ([Di Stasio et al., 2019](#)).
- **Hand-crafted physical indicators.** Swept-source OCT studies used the variability of the A-scan signal, a signal decay constant, and epithelial thickness, finding variability and decay to be reliable indicators of moderate dysplasia and carcinoma (summarised in the [2024 systematic review of OCT and AI for oral cancer](#)). Scattering-attenuation analysis of oral epithelial dysplasia was reported earlier by Tomlins et al. (2010).
- **Deep-learning segmentation.** The Vancouver group built a four-network pipeline (field-of-view detection, artifact detection, surface detection, and epithelial-stromal boundary detection using modified U-Nets) to produce en-face maps of epithelial depth across non-cancerous, precancerous, and cancerous pathologies ([Hill et al., 2024](#)). Building on it, they derived quantitative biomarkers from morphology, tissue attenuation, and layer contrast ([Malone et al., 2024](#)).
- **Label-efficient oral OCT.** Semi-supervised multi-task learning has been applied to oral OCT segmentation and denoising to cope with speckle and motion ([2025](#)).

Physics-informed OCT learning (other organs)

- **Attenuation estimation.** The depth-resolved single-scattering method estimates an attenuation coefficient for every pixel and was validated on layered phantoms with a 1300 nm system ([Vermeer et al., 2014](#)); layer-based refinements followed ([Cannon et al., 2021](#)).

- **Physics-informed networks.** QOCT-Net recovers pixel-level attenuation from intravascular OCT using a multiple-scattering model, trained on simulation and in vivo data ([2023](#)). PhysioSpeck-Net uses attenuation, speckle, and a retinal simulation model as auxiliary supervision for retinal OCT classification under dataset shift ([2026](#)). Deep scatterer-density estimation from speckle using a physically accurate noise model has also been shown ([2024](#)).

Trustworthy outputs

Conformal prediction produces intervals guaranteed to contain the true value with a user-chosen probability, without distributional or model assumptions ([Angelopoulos & Bates, 2021](#)), and has been extended to image-to-image regression in imaging ([Angelopoulos et al., 2022](#)). To our knowledge it has not been applied to layer-thickness measurement in oral OCT — to be confirmed.

Where the gap is

Capability	Oral OCT so far	This work
Epithelial segmentation	Supervised U-Net pipelines	Physics-informed, label-efficient
Attenuation	Hand-computed biomarker after segmentation	Built into learning as supervision and input representation
Labels needed	Full pixel annotations (or semi-supervised)	Very few, with physics and anatomical priors doing the rest
Pretraining source	Not studied	Physically matched (skin, cervix, oesophagus) vs abundant (retina, ImageNet)
Uncertainty	Not reported	Conformal intervals with guaranteed coverage; abstention
Cross-device behaviour	Single system	Tested explicitly

Positioning. The contribution is *not* “first to segment oral epithelium” or “first to use attenuation” — both exist. It is the **combination** of physics-informed self-supervision, label efficiency, physically matched transfer, and guaranteed measurement uncertainty, applied to oral soft tissue where data is scarcest.

3. Research questions

Primary research question

Can a learning framework that embeds the physics of OCT image formation measure oral mucosal epithelium with accuracy comparable to expert annotators using only a small fraction of the labelled data a conventional model needs, while providing measurement uncertainty with guaranteed coverage?

“Measure” here means locating the tissue surface and the epithelium–lamina propria boundary in every A-scan, and from them computing epithelial thickness (in micrometres, after refractive-index correction) and lamina-propria attenuation.

Sub-questions

- **RQ1 — Learning signal.** Does using a differentiable model of OCT signal formation as a training signal (re-rendering the scan from predicted tissue parameters) reduce the number of labels required, compared with supervised training and with generic self-supervised pretraining?
- **RQ2 — Representation.** Does giving the model physical tissue properties (depth-resolved attenuation maps) rather than raw intensity alone improve data efficiency and robustness to device-related variation?
- **RQ3 — Transfer source.** When pretraining on other data, does the *physical similarity* of the source (1300 nm OCT of epithelium over connective tissue: skin, cervix, oesophagus) matter more than its *size* (retinal OCT, natural images)? Does the best source depend on whether the oral site is keratinised (gingiva, palate) or not (cheek, floor of mouth)?
- **RQ4 — Trust.** Can conformal prediction wrap the thickness measurements in intervals that achieve their promised coverage and are narrow enough to be clinically informative, and does abstaining on low-quality scans lower error on the scans the system does answer?

Exploratory question

- **RQ5 — Speckle.** Do speckle statistics within the epithelium and lamina propria carry information about tissue state beyond layer geometry? (Pursued only if abnormal cases are available.)

4. Hypotheses

Each hypothesis states its rationale, a concrete prediction, how it is tested, and what result would refute it. “Label budget” means the fraction of the labelled training set used: 100%, 50%, 25%, 10%, 5%.

H1 — Physics-informed learning is label-efficient (answers RQ1)

- **Rationale.** Re-rendering a scan from predicted layer boundaries and attenuation forces the network to explain the image physically, so every unlabelled scan provides training signal.
- **Prediction.** At label budgets of 10% or less, the physics-informed model has lower boundary error than (a) nnU-Net trained on the same labels and (b) the same backbone pretrained with a generic masked autoencoder on the same unlabelled data.
- **Test.** Mean absolute boundary error (μm) and thickness error, patient-level cross-validation, paired comparison across patients.
- **Refuted if** the physics-informed model shows no statistically significant improvement over either baseline at any budget of 10% or less.

H2 — Attenuation representation improves efficiency and robustness (answers RQ2)

- **Rationale.** Raw brightness mixes tissue properties with device effects (focus, sensitivity roll-off); attenuation isolates the tissue.
- **Prediction.** Models given attenuation maps (alone or with intensity) reach a target accuracy with fewer labels, and degrade less when test scans are perturbed with realistic device effects or come from a second device.
- **Test.** Label-budget curves with and without attenuation input; performance drop under controlled perturbations and on external data.
- **Refuted if** attenuation input gives no consistent gain at low budgets and no smaller drop under perturbation.

H3 — Physical similarity beats data size for pretraining (answers RQ3)

- **Rationale.** Skin, cervical, and oesophageal OCT share both wavelength ($\sim 1300\text{ nm}$) and tissue architecture with oral mucosa; retinal OCT shares neither closely.

- **Prediction.** (a) At budgets of 10% or less, pretraining on physically matched sources outperforms retinal-OCT and ImageNet pretraining. (b) Skin pretraining helps more on keratinised sites (gingiva, palate); cervical/oesophageal pretraining helps more on non-keratinised sites. (c) A physics-based distance between domains (distributions of attenuation and speckle statistics) correlates negatively with transfer gain.
- **Test.** Same downstream task and budgets for each source; rank correlation between domain distance and gain.
- **Refuted if** retinal or ImageNet pretraining matches or beats matched sources, or no site-dependence or distance–gain relationship appears.
- **Dependency.** Requires access to at least two external source datasets; otherwise H3 is reduced to in-domain vs retinal vs ImageNet.

H4 — Guaranteed, useful uncertainty (answers RQ4)

- **Rationale.** Split conformal prediction gives finite-sample coverage guarantees with a small calibration set, suited to scarce data.
- **Prediction.** (a) 90% conformal intervals on epithelial thickness contain the expert value in at least 90% of held-out cases (within sampling error), per anatomical site. (b) Median interval width is smaller than the typical difference between healthy and abnormal thickness for that site. (c) Abstaining on the least-confident or low-quality scans reduces error on the remaining cases.
- **Test.** Empirical coverage and width per site; risk–coverage curve.
- **Refuted if** coverage falls clearly below nominal, or intervals are too wide to separate clinically distinct states.

H5 — Speckle carries extra signal (exploratory, answers RQ5)

- **Prediction.** Speckle-distribution parameters differ between healthy and abnormal tissue after controlling for thickness and site.
- **Refuted if** no significant difference remains after adjustment.

Why these hypotheses fit together

H1 and H2 are about **how** the model learns, H3 about **what it learns from**, and H4 about **whether its output can be trusted**. Together they answer the primary question; any one of H1, H2, or H4 confirmed on its own is already a publishable result.

5. Scope and variables

In scope. In vivo OCT of oral soft tissue (gingiva, labial and buccal mucosa, tongue, floor of mouth, palate); layer boundary localisation; epithelial thickness; lamina-propria attenuation; measurement uncertainty.

Out of scope for the first paper. Full diagnosis of dysplasia or cancer (too few abnormal cases expected); hard tissue (kept as optional second evaluation); polarization, angiography, and elastography (depend on device capability).

Type	Variables
Independent (what we change)	Learning signal (supervised / generic SSL / physics-informed); input representation (intensity / attenuation / both); pretraining source (none, ImageNet, retinal, skin, cervical, oesophageal, in-domain); label budget (100–5%)
Dependent (what we measure)	Boundary error (μm); thickness error and agreement with experts; Dice; robustness drop under perturbation; conformal coverage and interval width; error after abstention
Controlled (held fixed)	Backbone architecture and size; training schedule; patient-level data splits; annotation protocol; refractive-index correction; evaluation code
Stratifying factors	Anatomical site; keratinised vs non-keratinised; image quality; device (if more than one)
Known confounders to record	Probe contact pressure (soft tissue compresses), motion, saliva film, focus position, operator

6. Expected contributions

1. **A physics-informed, label-efficient method** for oral OCT layer measurement that learns from unlabelled scans by re-rendering them through a differentiable model of OCT signal formation.
2. **Evidence on representation and transfer:** whether attenuation maps and physically matched pretraining sources beat raw intensity and abundant-but-mismatched sources when labels are scarce — including a site-dependent (keratinised vs non-

keratinised) analysis.

3. **Guaranteed measurement uncertainty** for oral epithelial thickness via conformal prediction, with an abstention mechanism — the measurement layer of a clinical decision support system.
4. **A benchmark protocol** for oral OCT under label scarcity (label-budget curves, patient-level splits, inter-annotator ceiling, device-perturbation tests) that others can reuse.
5. **Optionally, a released dataset** (de-identified, if ethics and agreements allow) — valuable in itself given that no public oral OCT dataset exists.

7. Assumptions, risks, and what must be confirmed

Item	Why it matters	Mitigation or check
Access to an OCT device and in vivo soft-tissue scans	Everything depends on target data	Confirm with project lead; healthy-volunteer collection is the fastest route
Raw (linear, unprocessed) data export	Attenuation and physics-based learning need it	Check device software; request from collaborators explicitly
Enough unlabelled scans for self-supervision	H1-H3 rely on unlabelled volume	Even ~30 subjects × several sites yields thousands of B-scans (correlated, so report per patient)
Two or more annotators	Inter-annotator variability is the realistic accuracy ceiling	Annotate a shared subset with a written protocol
External source datasets (skin, cervix, oesophagus)	Needed for full H3	Reduce H3 scope if unavailable
Single-scattering model is too simple	Physics model may misfit deep tissue, rete ridges, vessels	Add a learned residual; restrict to depths with adequate signal
Probe compression of soft tissue	Changes measured thickness	Record pressure/contact where possible; stratify or standardise protocol

Item	Why it matters	Mitigation or check
Novelty	Physics-informed OCT learning exists in retina and intravascular OCT; oral segmentation exists	Targeted literature check before writing; position as the combination listed in Section 2
Few abnormal cases	Limits clinical claims	Keep diagnosis out of scope; H5 exploratory only

8. References

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